

UCE856 :ADVANCED SURVEYING				
	L	T	P	Cr
	3	0	2*	3.5
Course Objective: Apply the knowledge of modern surveying techniques like remote sensing and GIS, aerial surveying and Drone surveying for various projects, which is an essential tool for any civil engineering projects				
Global Positioning Systems (GPS): Working Principle, Types of GPS, Application of GPS, DGPS-working Principle, DGPS errors (RT & PP).				
Fundamentals of Remote Sensing: Introduction to Remote sensing and electromagnetic spectrum, types of sensors, Orbit and path, Geometric and Radiometric errors, image interpretation-visual and digital				
GIS techniques: Basic concepts, Topology, Data models- Vector and Raster, attribute, Digital Elevation models, Map, Projections				
Aerial Surveying and Photogrammetry: Basic concepts, principles, and applications of photogrammetry, concepts, and applications for map preparation, Stereoscopy, LIDAR				
Drone Surveying: Introduction to drones, comparison of surveying drone and its accuracy; techniques of controlling errors				
Course Learning Objectives (CLO) The students will be able to: 1. Process the raw satellite imagery and prepare landuse and land cover maps and integrated with GIS 2. Process aerial survey data and estimate vertical and horizontal distances. 3. Carry out drone survey and process the data for various uses.				
Text Books: 1. Gopi, S., Sathikumar, R, and Madhu, N. Advanced Surveying, Pearson Publisher, 2023 2. Garg, P.K., Introduction to surveying and geomatics engineering, CBS Publishing, 2023 3. Arora KR, Surveying (Vol. III). Standard Book House, 2015				

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weights (%)
1.	MST	30
2.	EST	40
3.	Sessionals (May include Assignments/Projects/Quizzes/Tutorials/Lab Evaluations)	30